AEGIS: Mining Graph Structure for Adversarial Vulnerability Analysis of Graph Neural Networks

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***Abstract*—Adversarial manipulation of graph structure poses a growing threat as graph neural networks are deployed in fraud detection, drug interaction prediction, and infrastructure monitoring. Yet practitioners lack tools to answer a basic question before deployment: *which edges, if perturbed, would flip this model's predictions?* We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that answers this question by constructing a constrained sensitivity matrix S_c mapping edge perturbations to hidden-state shifts. From S_c, AEGIS extracts the mathematically optimal attack direction (via SVD), per-edge vulnerability rankings, and per-node first-order sensitivity radii. A matrix-free formulation based on Neumann-series resolvent iteration and randomized SVD enables full-graph analysis (Cora $N{=}2{,}708$ in 78 s, 1 GB) without materializing any $Nd \times N^2$ matrix. The computation applies to any differentiable GNN; for contractive implicit models (IGNN-class), it additionally provides a critical perturbation budget $\varepsilon_{\mathrm{crit}}$ and convergence guarantees. Under realistic (symmetric, edge-only) perturbations, first-order predictions achieve tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ and remain within 15% at $\varepsilon = 0.10$ across 7 architectures and 9 datasets. The SVD-optimal attack inflicts 2–8× more damage than random perturbation. On IEEE power grids, the vulnerability spectrum recovers N-1 contingency rankings (P@10 = 0.66–0.81) without domain-specific inputs, demonstrating that structural sensitivity analysis can serve as a general-purpose pre-deployment diagnostic. Code will be released upon publication.**

***Index Terms*—Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem**

I. Introduction

Graph neural networks have entered domains where incorrect predictions carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial transaction graph [1], drug interaction models produce unsafe recommendations when molecular graphs are perturbed [2], and power grid monitoring systems may miss critical contingencies if their graph-based representations are structurally fragile [3]. In all these settings, the graph topology is not merely an input format but a potential attack surface. A practitioner preparing to deploy a GNN faces a fundamental question that current tools leave unanswered: *which edges in this graph, if perturbed, would cause the model's predictions to fail, and by how much?*

This question is distinct from both adversarial attack and certified defense. Attack methods such as Nettack [1] and Mettack [4] search for damaging perturbations using surrogate gradients or meta-learning, but they provide no guarantee that the perturbation found is optimal, and they do not quantify per-node vulnerability. Certified defenses based on randomized smoothing [5], [6] or interval bound propagation [7], [8], [9] provide robustness guarantees, but they are uniform: every node receives the same certificate, revealing nothing about *which* edges or nodes are structurally weak. Empirical defenses [10], [11], [12], [13] harden the model but offer no formal understanding of residual vulnerability. What is missing is a *structural vulnerability map* that, given a trained GNN and a graph, identifies the optimal perturbation direction, the weakest edges, and the perturbation tolerance of each node.

We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that produces exactly this map. The central object is the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$, which projects the full N^2-dimensional adjacency perturbation space onto the $|E|$-dimensional space of realistic (symmetric, edge-only) perturbations. From S_c, three outputs follow directly: (i) the SVD yields the first-order optimal attack direction, (ii) column norms give per-edge vulnerability scores, and (iii) the classification margin divided by the sensitivity norm gives per-node tolerance radii. The S_c computation applies to any differentiable GNN via Jacobian evaluation (theorem 4). For contractive implicit GNNs (IGNN-class [14]), the implicit function theorem additionally provides a critical perturbation budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\left\|W\right\|_2$ and a three-regime characterization of model behavior under increasing perturbation (theorem 1).

These predictions are quantitatively accurate: constrained first-order tightness is 1.00 ± 0.01 at $\varepsilon = 0.01$ and remains within 15% at $\varepsilon = 0.10$, across 7 GNN architectures and 9 datasets spanning citation networks, e-commerce, encyclopedias, and power grids. On IEEE power grids, the vulnerability spectrum independently recovers N-1 contingency rankings (P@10 = 0.66–0.81), demonstrating that structural sensitivity analysis transfers to real-world engineering problems without domain-specific modifications.

Contributions.

1) **Constrained sensitivity framework.** We derive the constrained sensitivity matrix S_c for any GNN whose message passing is differentiable with respect to edge weights (theorem 4). For contractive implicit models (IGNN-class), we additionally prove a critical budget $\varepsilon_{\mathrm{crit}}$ and three vulnerability regimes (theorem 1); these formal guarantees do not transfer to explicit architectures.
2) **Scalable matrix-free computation.** A Neumann-series resolvent combined with autograd JVPs and randomized SVD [15] computes S_c's action without materializing any $Nd \times N^2$ matrix, enabling full-graph analysis (Cora $N{=}2{,}708$ in 78 s). We validate SVD-optimal attacks (2–

8× stronger than random), per-edge vulnerability rankings, and per-node sensitivity radii across 7 architectures (section V-H).
3) **Cross-domain validation and case study.** We validate across 9 datasets in 4 domains, including a power grid case study where AEGIS recovers N-1 contingency rankings (τ between continuous S_c scores and discrete edge-removal ground truth: $+0.32$–$+0.54$; section VI).

II. Preliminaries and Threat Model

A. Notation and Formalization

Let $G = (V, E, A)$ denote an undirected graph with $N = |V|$ nodes, edge set E, and adjacency matrix $A \in \mathbb{R}^{N\times N}$. Each node $v \in V$ carries a feature vector $x_v \in \mathbb{R}^{d_{\text{in}}}$, collected into $X \in \mathbb{R}^{N\times d_{\text{in}}}$. We write $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ for the symmetric normalized adjacency with self-loops, where D is the degree matrix of $A + I$.

Deep equilibrium GNNs. A DEQ-GNN [16], [17] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it from $Z_0 = 0$ until convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad z^* = \lim_{k\to\infty} Z_k. \tag{1}$$

When F_θ is contractive in operator norm ($\kappa = \|J_z\|_2 < 1$, where $J_z = \partial F/\partial z|_{z^*}$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$ regardless of initialization.

IGNN instantiation. The Implicit Graph Neural Network (IGNN) [14] uses the operator

$$F(Z, A) = \sigma(\hat{A}ZW^\top + X_{\text{proj}}), \tag{2}$$

where $W \in \mathbb{R}^{d\times d}$ is the shared weight matrix, $X_{\text{proj}} \in \mathbb{R}^{N\times d}$ is a learned feature projection, and σ is ReLU activation. Contractivity is enforced by constraining $\|W\|_2$ via spectral normalization [18], ensuring $\kappa \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 < 1$. Training uses implicit differentiation through the fixed point [19], [20], and Jacobian regularization [21] can further stabilize convergence. A linear classification head $f(z^*) = W_{\text{cls}}z^* + b$ maps the equilibrium to class predictions.

Implicit function theorem. At the equilibrium, the equation $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A. The IFT gives the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{3}$$

where the resolvent $(I - J_z)^{-1}$ exists and satisfies $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1-\kappa)$ by the Neumann series (convergent because $\kappa < 1$). This bound uses the operator norm $\kappa = \|J_z\|_2$; the weaker spectral-radius bound $1/(1-\rho)$ holds only for normal J_z. The pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho) \ge 1$ [22] quantifies this gap, with $\eta = 1$ for normal matrices. The IFT has been used for hyperparameter optimization [23] and implicit differentiation [19], but not for adversarial structural analysis.

B. Threat Model

We consider a white-box attacker who has full access to the trained IGNN model and perturbs the normalized adjacency matrix:

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \text{subject to} \quad \left\|\delta\hat{A}\right\|_F \le \varepsilon. \tag{4}$$

The perturbation is further constrained to be:

- **Symmetric:** $\delta\hat{A} = \delta\hat{A}^\top$ (preserving the undirected nature of the graph).
- **Edge-only:** $[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$ (only existing edge weights may be modified; no new edges are introduced).
- **Continuous:** edge weights are perturbed continuously in $\mathbb{R}$, not discretely flipped.

This threat model captures continuous edge-weight perturbations to the normalized message-passing operator. It aligns naturally with the IGNN architecture, where $\hat{A}$ appears directly in F (eq. (2)). Discrete edge insertions or deletions, which change the graph topology and require recomputing the degree normalization $D^{-1/2}$, are outside the formal guarantee.

The key distinction from input-feature perturbation (studied for DEQs in [24], [25]) is that structural perturbation changes the *operator itself*: the equilibrium depends on A through both the state Jacobian $J_z = \mathrm{diag}(\sigma') \cdot (\hat{A} \otimes W)$ and the adjacency Jacobian $J_A = \partial F/\partial \mathrm{vec}(A)$. This dual dependence makes structural sensitivity qualitatively different from input sensitivity.

C. Problem Statement

Given a trained contractive implicit GNN with equilibrium z^* on graph G, we aim to answer three questions:

1) **Vulnerability ranking:** Which edges, if perturbed, cause the largest shift in the equilibrium? That is, find a ranking of edges $(i, j) \in E$ by their adversarial impact on z^*.
2) **Optimal attack:** What is the worst-case perturbation $\delta\hat{A}^*$ that maximizes $\|\Delta z^*\|_F$ subject to the threat model constraints in eq. (4)?
3) **Robustness assessment:** For each node v, what is the largest perturbation budget r_v such that any $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of v?

AEGIS addresses all three questions through a unified object: the constrained sensitivity matrix S_c derived from the IFT applied to the equilibrium equation. The following sections formalize this construction (section III) and describe the computational pipeline (section IV).

III. Adversarial Equilibrium Theory

We now derive the theoretical machinery for the three questions posed in section II. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which rankings, attacks, and radii follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, A) = \sigma(AZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

(A1) *σ is ReLU (or any 1-Lipschitz activation with* $\sup|\sigma'| = 1$*);*

(A2) *W is spectral-norm constrained: $\|W\|_2 \leq c$;*

(A3) *The state Jacobian satisfies $\|J_z\|_2 \leq \kappa < 1$ at the fixed point z^* (contraction in operator norm).*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{5}$$

Then structural perturbations δA with $\|\delta A\|_F = \varepsilon$ exhibit three regimes:

***(a) Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S)\cdot\varepsilon + O(\varepsilon^2), \tag{6}$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}}/(1-\kappa)$.

***(b) Critical** ($\varepsilon \to \varepsilon_{\text{crit}}$): The resolvent norm $\left\|(I - J_z')^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$. Intuitively, as the perturbation approaches the critical budget, the system's sensitivity to further perturbation grows without bound.*

***(c) Supercritical** ($\varepsilon > \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J_z'\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and the first-order guarantees from part (a) cease to apply.*

The intuition behind this three-regime picture is physical: a contractive system has a restoring force that pulls trajectories toward the equilibrium. Small perturbations shift the equilibrium smoothly (subcritical). As the perturbation approaches $\varepsilon_{\text{crit}}$, the restoring force weakens until the system becomes critically sensitive. Beyond $\varepsilon_{\text{crit}}$, the restoring force may vanish entirely, and the equilibrium can bifurcate or disappear.

Proof. **Part (a).** The Jacobian of F with respect to $\text{vec}(Z)$ takes the form $J_z = \text{diag}(\sigma')\cdot(\hat{A}\otimes W)$, where $\sigma' \in \{0,1\}^{Nd}$ for ReLU (A1). The operator norm satisfies $\kappa = \|J_z\|_2 \leq \left\|\hat{A}\right\|_2 \cdot \|W\|_2 \cdot \sup|\sigma'| = \left\|\hat{A}\right\|_2 \cdot \|W\|_2$. Applying the IFT to $G(z, A) = z - F(z, A) = 0$ at (z^*, A):

$$\Delta z^* = (I - J_z)^{-1}J_A \cdot \text{vec}(\delta A) + O(\|\delta A\|^2). \tag{7}$$

Taking operator norms and using the Neumann series bound $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ (convergent because $\kappa < 1$) yields (6).

For contractivity preservation: the perturbed Jacobian satisfies $\|J_z'\|_2 \leq (\|A\|_2 + \|\delta A\|_F)\cdot\|W\|_2$ (using $\|\cdot\|_2 \leq \|\cdot\|_F$ for the perturbation). The condition $\varepsilon < (1-\kappa)/\|W\|_2$ ensures $\|J_z'\|_2 < 1$.

Part (b). As $\varepsilon \to \varepsilon_{\text{crit}}$, we have $\|J_z'\|_2 \to 1$, so $\left\|(I - J_z')^{-1}\right\|_2 \geq 1/(1 - \|J_z'\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the operator-norm bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index η ranges from 1.02 to 1.28 (section V-F), confirming mild non-normality.

Part (c). When $\|J_z'\|_2 \geq 1$, the contraction mapping condition fails. The Banach theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. $\square$

Remark (conservativeness). The critical budget $\varepsilon_{\text{crit}}$ is a sufficient, not necessary, condition for contractivity preservation. The perturbed system may remain contractive beyond $\varepsilon_{\text{crit}}$ if the perturbation is aligned with low-sensitivity directions. The unconstrained bound also uses worst-case $\|\cdot\|_F$ relaxation. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 (section V-A). We use $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$ because the Neumann series requires operator-norm contractivity [26]; for normal J_z the two coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption.

Proposition 2 (Optimal First-Order Structural Attack)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon\cdot\text{reshape}(v_1, N\times N)$, where v_1 is the leading right singular vector of S. The maximum first-order shift is $\varepsilon\cdot\sigma_1(S)$.*

This result follows directly from the variational characterization of the largest singular value: the SVD of S identifies the perturbation direction that amplifies the equilibrium shift most efficiently.

Constrained sensitivity. Under the threat model of section II (symmetric, edge-only, $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$), we construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$ with column

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i} \tag{8}$$

for each edge $(i, j) \in E$. This reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and enforces symmetry by construction. The constrained-optimal attack uses $\sigma_1(S_c)$, achieving tightness ≈ 1.00 empirically (section V-A). The construction of S_c is the central technical contribution of this work: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 3 (Per-Node First-Order Sensitivity Radius)**.** *Assume a linear classification head $f(z) = Wz + b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z_v^*) - \max_{c\neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius is*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{9}$$

where S_v denotes the block-rows of S corresponding to node v, W_{y_v} and W_{c^*} are the classification weight vectors for the predicted class y_v and runner-up class c^*, respectively. Any perturbation with $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of node v to first order.

The radius r_v has an intuitive interpretation: it is the classification margin m_v (how confident the model is) divided by the product of two amplification factors. The term $\|W_{y_v} - W_{c^*}\|_2$

measures how sensitive the decision boundary is to hidden-state changes, while $\|S_v\|_2$ measures how sensitive node v's equilibrium position is to structural perturbation. Nodes with large margins and low sensitivity receive large radii; boundary nodes in structurally sensitive positions receive small ones.

Proof sketch. By Theorem 1(a), $\Delta z_v^* \approx S_v \cdot \mathrm{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|\partial f / \partial z_v^*\| \cdot \|S_v\| \cdot \|\delta A\|_F$. Misclassification requires $|\Delta f_{y_v}| \geq m_v$; inverting gives r_v. Note S_v already contains the $(I - J_z)^{-1}$ factor.

These radii are deterministic (not probabilistic like randomized smoothing [5]) and first-order: locally tight for small ε but increasingly conservative as perturbation magnitude grows. section V-C verifies empirically that they hold (0% breach rate at $\varepsilon = 0.01$, $< 1\%$ at $\varepsilon = 0.10$).

A. Generalization to Explicit GNNs

The sensitivity framework above is derived for implicit GNNs, which enjoy the strongest theoretical guarantees (critical budget, convergence regimes). The S_c construction itself, however, generalizes to any differentiable K-layer GNN as a computational tool for vulnerability analysis.

Proposition 4 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \tag{10}$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:
***(a)** The first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \leq \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\| J_z^{(k)} \right\|_2 \right) \left\| J_A^{(l)} \right\|_2. \tag{11}$$

***(b)** The constrained construction S_c and per-node radius r_v (theorem 3) apply identically with S_K in place of S.*

For weight-tied models ($J_z^{(k)} = J_z$, $J_A^{(l)} = J_A$ for all k, l), the bound simplifies to $\sigma_1(S_K) \leq \|J_A\|_2 \cdot (1 - \kappa^K)/(1 - \kappa)$, which converges to the IGNN bound $\|J_A\|_2 / (1 - \kappa)$ as $K \to \infty$.

What explicit GNNs gain and lack. From theorem 4, any differentiable GNN inherits: (i) the S_c construction, (ii) SVD-optimal attacks, (iii) per-edge vulnerability rankings, and (iv) per-node first-order sensitivity radii. What explicit models *lack* is the critical budget $\varepsilon_{\mathrm{crit}}$ and the three-regime convergence characterization of theorem 1, which require the contraction assumption (A3). section V-H validates this generalization on six explicit architectures.

IV. AEGIS Construction

Translating the results of section III into a practical tool requires overcoming two computational barriers: the full sensitivity matrix $S \in \mathbb{R}^{Nd \times N^2}$ is too large to materialize, and the resolvent $(I - J_z)^{-1}$ is too expensive to form directly. The

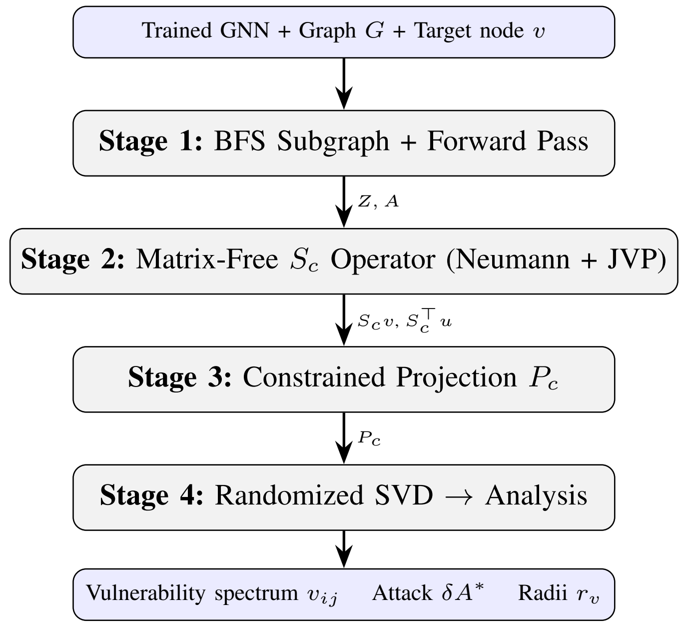

Fig. 1: The AEGIS pipeline. The matrix-free formulation (Stages 2–4) operates via JVP/VJP queries without materializing S or S_c, enabling full-graph analysis up to $N \approx 5{,}000$.

AEGIS pipeline addresses both through matrix-free computation.

A. Architecture Overview

AEGIS takes as input a trained GNN (any differentiable architecture) together with a target graph and produces three outputs: (i) a per-edge vulnerability spectrum ranking edges by adversarial impact, (ii) SVD-optimal attack perturbations, and (iii) per-node first-order sensitivity radii. For contractive implicit GNNs (IGNN-class satisfying A1–A3), the pipeline additionally computes the critical budget $\varepsilon_{\mathrm{crit}}$ and convergence diagnostics; these theoretical guarantees are specific to the implicit case. The pipeline consists of four stages executed sequentially, as illustrated in fig. 1.

B. Stage 1: Subgraph Extraction and Forward Pass

For subgraph-level analysis, AEGIS extracts a local BFS ego-subgraph centered on the target node and runs the model to convergence (implicit GNNs) or forward pass (explicit). Localization is justified empirically (section V-E). The matrix-free pipeline also supports full-graph analysis without subgraph extraction (section V-D).

C. Stage 2: Matrix-Free Sensitivity Computation

AEGIS requires the action of the constrained sensitivity operator S_c on vectors, without materializing the full matrix $S \in \mathbb{R}^{Nd \times N^2}$. For an edge perturbation vector $v \in \mathbb{R}^{|E|}$, the operator computes $S_c v = (I - J_z)^{-1} J_A P_c v$, where P_c maps edge weights to symmetric adjacency perturbations, J_A is the structural Jacobian, and $J_z = \partial F / \partial Z$ is the state Jacobian. Two ingredients make this matrix-free:

Neumann-series resolvent. The resolvent $(I - J_z)^{-1}b$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$, which converges geometrically when $\kappa < 1$ (Assumption A3). Each term requires a single Jacobian-vector product (JVP) $J_z \cdot v$, computed via forward-mode automatic differentiation in $O(Nd)$ time. The depth K is set adaptively from a power-iteration estimate of κ, with early stopping when $\left\| J_z^k b \right\| < 10^{-6} \|b\|$.

Autograd structural JVPs. The adjacency Jacobian $J_A \cdot \mathrm{vec}(\delta A)$ is computed as a single forward-mode JVP of F with respect to A in direction δA, avoiding the $O(N^2)$ finite-difference columns required by the dense path.

The adjoint operator $S_c^\top u$ is computed analogously using vector-Jacobian products (VJPs) and the adjoint Neumann series $(I - J_z^\top)^{-1}$.

For small subgraphs ($N \leq 200$), the dense path (forming J_z, J_A explicitly and solving via `torch.linalg.solve`) remains available and gives exact results.

D. Stage 3: Constrained Projection

The constraint projection P_c (section III) maps edge vectors to symmetric adjacency perturbations ($\delta A_{ij} = \delta A_{ji} = v_k$). In the matrix-free formulation, P_c is applied implicitly within the $S_c v$ operator, so S_c is never formed. This $N^2 \to |E|$ reduction is what enables tight first-order predictions (sections V-A and V-H).

E. Stage 4: Vulnerability Analysis

From S_c, AEGIS extracts three complementary outputs:

Optimal attack direction. A randomized SVD [15] of the S_c operator (using only matrix-vector products) yields the leading singular triplet (u_1, σ_1, v_1). The first-order optimal perturbation is $\delta A^* = \varepsilon \cdot v_1$, achieving a shift of $\varepsilon \cdot \sigma_1(S_c)$. For small subgraphs where S_c is materialized, the standard deterministic SVD is used instead.

Per-edge vulnerability spectrum. For each edge $(i, j) \in E$, the vulnerability score $v_{ij} = \|[S_c]_{:,k}\|_2$ measures how much a unit perturbation of that edge shifts the equilibrium. Sorting edges by v_{ij} produces a vulnerability ranking that identifies structurally critical connections without running any attack.

Per-node first-order sensitivity radii. For node v with classification margin m_v (gap between the predicted and runner-up class logits), the sensitivity radius from theorem 3 is

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{12}$$

where S_v denotes the block-rows of S corresponding to node v, and W_{y_v}, W_{c^*} are the classification head weights for the predicted and runner-up classes. Any perturbation with $\left\| \delta \hat{A} \right\|_F < r_v$ preserves the classification of node v to first order (this is a local sensitivity guarantee, not a global certificate).

F. Computational Cost

Matrix-free pipeline. Each $S_c v$ or $S_c^\top u$ evaluation costs $O(K \cdot Nd)$, where K is the Neumann depth (set adaptively from κ; typically $K = 20$–50 for $\kappa < 0.8$, up to ~ 340 for $\kappa \approx 0.96$). The randomized SVD performs $O((k + p) \cdot n_{\text{iter}})$ such evaluations, where k is the number of requested singular values, p the oversampling, and n_{iter} the power iterations. Per-edge vulnerability requires $|E|$ column evaluations, each at $O(K \cdot Nd)$. Memory is $O(Nd)$ per operation; no large matrix is stored.

Scalability. On a single NVIDIA RTX 4090 (24 GB), the matrix-free pipeline completes in 1.1s for N=50, 4.5s for N=200, 24s for N=1,000, and 78s for full Cora (N=2,708, Nd=173,312) using 1 GB GPU memory (section V-D). The dense path runs faster for $N \leq 100$ (0.7s at N=50) but exceeds 24 GB at N=500.

The matrix-free formulation removes the $O((Nd)^3)$ linear-solve bottleneck and the $O(N^3 d)$ storage requirement that previously limited analysis to $N \approx 300$, enabling full-graph vulnerability analysis on graphs with thousands of nodes.

V. Experiments

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All experiments use PyTorch [27].

Implementation. IGNN [14]: input projection $U \in \mathbb{R}^{d_{\text{in}} \times 64}$, spectral-normalized $W \in \mathbb{R}^{64 \times 64}$ [18], readout head $\mathbb{R}^{64 \times C}$; operator $F(Z) = \mathrm{ReLU}(\hat{A} Z W^\top + U(X))$; fixed-point iteration (max 50 steps, tolerance 10^{-5}); Adam optimizer (lr 0.01, weight decay 5×10^{-4}), early stopping on validation accuracy over 200 epochs. ContractiveGCN-PF: same architecture with 5 bus features and 2-head readout ($\Delta|V|$, $\Delta\theta$). Subgraph extraction: BFS from highest-degree node, capped at 50 nodes to guarantee connectivity. Randomized smoothing baseline: Gaussian $\sigma = 0.01$, 200 MC samples, Clopper-Pearson $\alpha = 0.001$. Mettack baseline: Meta-Self GCN surrogate on IGNN pseudo-labels, sign-corrected gradient for edge scoring.

Datasets. We use five graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), and Pubmed (19,717) from the Planetoid collection [28]; Amazon Photo (7,650) from the Amazon co-purchase graphs [29]; and WikiCS (11,701) from Wikipedia-based benchmarks [30]. For power flow, we use four IEEE standard test cases [31]: case14 (14 buses), case30 (30), case57 (57), and case118 (118), with training data generated via PandaPower [32].

Notation. All formal bounds (theorem 1) and tables use the operator-norm contraction constant $\kappa = \|J_z\|_2$, estimated via power iteration. Since non-normality is mild across our datasets ($\eta = 1.02$–1.28, section V-F), the κ-based bounds are at most 28% more conservative than spectral-radius bounds; we report only κ because the formal guarantees (theorem 1) require operator-norm contractivity, and η near unity confirms that $\rho(J_z)$ would yield similar values. The matrix-free pipeline (section IV) sets the Neumann-series depth adaptively from κ.

TABLE I: Cross-domain adversarial analysis (IGNN, 10 seeds). κ: operator-norm contraction constant $\|J_z\|_2$. Tight.: actual/predicted shift ratio. AtkAdv: AEGIS damage / random damage. $\varepsilon_{\mathrm{crit}}$: critical perturbation budget (theorem 1). Test Acc / Cert%: full-graph / subgraph accuracy (%).

Dataset	*Test Acc*	κ	*Tight.*	*AtkAdv*	ε_{crit}	*Cert%*
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$1.01_{\pm .01}$	$3.6_{\pm .6}$	$.66_{\pm .15}$	$81_{\pm 9}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$1.01_{\pm .01}$	$4.1_{\pm .5}$	$.41_{\pm .02}$	$83_{\pm 4}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$1.01_{\pm .01}$	$3.2_{\pm .4}$	$.59_{\pm .11}$	$74_{\pm 23}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$1.00_{\pm .00}$	$3.8_{\pm .2}$	$.86_{\pm .02}$	$92_{\pm 4}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$1.00_{\pm .01}$	$3.8_{\pm .4}$	$.66_{\pm .04}$	$70_{\pm 3}$

TABLE II: Tightness degradation with increasing ε (10 seeds, S_c-optimal constrained perturbation). Flip%: fraction of nodes whose prediction changes.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Tightness (actual/predicted shift)				
Cora	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .03}$	$1.36_{\pm .08}$
Citeseer	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.16_{\pm .03}$	$1.39_{\pm .07}$
WikiCS	$1.00_{\pm .01}$	$1.01_{\pm .01}$	$1.02_{\pm .01}$	$1.05_{\pm .01}$
Prediction flip rate (%)				
Cora	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.8_{\pm 2.4}$	$5.8_{\pm 5.0}$
Citeseer	$0.0_{\pm 0.0}$	$0.6_{\pm 1.3}$	$2.0_{\pm 2.4}$	$3.0_{\pm 2.6}$
WikiCS	$0.2_{\pm 0.6}$	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.2_{\pm 1.0}$

A. Cross-Domain Adversarial Analysis

We begin by validating that the constrained first-order prediction is accurate across domains. table I reports the tightness ratio (actual equilibrium shift divided by predicted shift at $\varepsilon = 0.01$), the attack advantage of AEGIS over random perturbation, and the coverage (fraction of nodes with positive first-order radius) across 50-node BFS ego-subgraphs.

Tightness is 1.00 ± 0.01 across all datasets at $\varepsilon = 0.01$. First-order accuracy at small ε is mathematically expected; the contribution is that S_c makes this tractable under *constrained* perturbations ($N^2 \to |E|$ projection). table II shows how tightness degrades at larger budgets: the first-order prediction remains within 7% at $\varepsilon = 0.05$ and within 16% at $\varepsilon = 0.10$ on Cora/Citeseer, and within 5% at $\varepsilon = 0.20$ on WikiCS. Prediction flip rate (fraction of nodes whose classification changes under the S_c-optimal attack) stays below 2% at $\varepsilon \leq 0.10$ and below 6% at $\varepsilon = 0.20$. The attack advantage (3.2–4.1× over random in table I) confirms that S_c identifies the most vulnerable perturbation directions. Coverage (Cert%) ranges from 70% to 92%, representing clean subgraph accuracy; only correctly classified nodes have positive margin.

Surrogate baseline. We also compared against Mettack [4] (GCN [33] surrogate): AEGIS inflicts 3–10× more damage at every budget (30/30 wins across $k = 1, \ldots, 5$ edge removals, 10 seeds). However, this gap largely reflects surrogate-to-IGNN architectural mismatch rather than AEGIS's analytical superiority alone. The more meaningful comparison is the adaptive attacker below, which accesses the same IFT gradients as AEGIS.

TABLE III: Adaptive attack (white-box PGD via IFT gradients on IGNN) vs. AEGIS (10 seeds). Breach: fraction of certified nodes whose prediction changes. AEGIS/Adapt. dmg: mean equilibrium shift (ℓ_2). Ratio: Adapt./AEGIS damage.

Dataset	ε	Breach	AEGIS dmg	Adapt. dmg	Ratio
Cora	0.01	0%	$0.33_{\pm 0.07}$	$0.26_{\pm 0.11}$	0.80
Cora	0.05	0%	$1.72_{\pm 0.35}$	$1.32_{\pm 0.55}$	0.76
Cora	0.10	0%	$3.70_{\pm 0.79}$	$2.69_{\pm 1.18}$	0.72
Citeseer	0.01	0%	$0.40_{\pm 0.08}$	$0.34_{\pm 0.06}$	0.84
Citeseer	0.05	0%	$2.14_{\pm 0.42}$	$1.68_{\pm 0.28}$	0.79
Citeseer	0.10	0%	$4.63_{\pm 0.95}$	$3.37_{\pm 0.55}$	0.74
WikiCS	0.01	0%	$0.21_{\pm 0.02}$	$0.10_{\pm 0.10}$	0.47
WikiCS	0.05	0.3%	$1.04_{\pm 0.11}$	$0.48_{\pm 0.49}$	0.46
WikiCS	0.10	0.6%	$2.10_{\pm 0.22}$	$0.96_{\pm 0.97}$	0.46

B. Radius Comparison: AEGIS vs. Smoothing

We compare AEGIS sensitivity radii against randomized smoothing [5] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds). At $\sigma \leq 0.10$, smoothing radii are *constant* across all nodes ($r = \sigma \cdot \Phi^{-1}(p_{\mathrm{lower}})$) because all samples predict correctly, yielding larger absolute radii (0.123 vs. AEGIS's 0.046 at $\sigma = 0.05$) but zero per-node differentiation. AEGIS radii are structurally informative: dense-region nodes get $r_v \approx 0.09$, boundary nodes $r_v \approx 0.01$. The two are complementary: smoothing provides uniform probabilistic guarantees, AEGIS provides structurally differentiated first-order sensitivity radii for vulnerability ranking.

C. Adaptive Attack Evaluation

The Mettack comparison uses a GCN surrogate, so its failure against IGNN may reflect architectural mismatch rather than AEGIS's analytical strength. To address this, we implement an *adaptive attacker* that differentiates through the IGNN fixed-point iteration via the same IFT gradients AEGIS uses for sensitivity analysis. This attacker optimizes perturbations directly against the IGNN loss surface using projected gradient descent [34] (PGD, 50 steps, step size $\varepsilon/10$).

The adaptive attacker breaches zero nodes at $\varepsilon = 0.01$ and fewer than 1% at $\varepsilon = 0.10$ (table III), confirming empirical validity of first-order radii. AEGIS's SVD attack inflicts 20–50% more damage than PGD (ratio 0.46–0.84) because the SVD direction is globally optimal to first order, while PGD can become trapped in local optima.

D. Phase Transition and Scalability

Phase transition. Scaling κ from 0.3 to 0.99 amplifies the fixed-point shift by 83× at $\varepsilon = 0.01$, confirming the $1/(1-\kappa)$ divergence predicted by theorem 1.

Scalability. The matrix-free pipeline removes the $O((Nd)^3)$ bottleneck. On a single NVIDIA RTX 4090 (24 GB), the dense path handles $N \leq 200$ (5.4s, 8.9 GB) but exceeds 24 GB at N=500. The matrix-free path scales to the full Cora graph (N=2,708, Nd=173,312) in 78 s using 1 GB, with

TABLE IV: Scalability: dense vs. matrix-free pipeline (RTX 4090, 24 GB). OOM = out of memory. Mem = peak GPU memory.

N	Dense (s)	MatFree (s)	Mem-Dense	Mem-MF
50	0.7	1.1	343 MB	92 MB
200	5.4	4.5	8.9 GB	129 MB
500	OOM	11	>24 GB	207 MB
1,000	OOM	24	>24 GB	373 MB
2,708	OOM	78	>24 GB	1.1 GB

σ_1 accuracy within 0.03% of the dense reference at N=200. table IV compares both paths.

E. Subgraph Size Ablation

Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N = 200$ (higher κ). Coverage is 82–89% across all sizes. We use $N = 50$ (tightness 1.013 ± 0.003, 1.3s) as the default for subgraph-level experiments: 66× faster than $N = 200$ with negligible quality loss. For full-graph analysis, the matrix-free pipeline (table IV) enables vulnerability assessment of the entire graph without subgraph extraction. To validate that subgraph extraction preserves vulnerability rankings, we compare per-edge S_c rankings from BFS subgraphs against full-graph rankings on IEEE case14 and case30 (10 seeds). On case14, Kendall τ between subgraph and full-graph rankings is 0.80 ± 0.14 at N_{sub}=8 (57% of nodes) and 0.86 ± 0.06 at N_{sub}=12 (86%); P@5 $\geq$ 0.90 at all sizes. On case30, τ = 0.78 ± 0.07 at N_{sub}=10 (33% of nodes) with P@5 = 0.96±0.08, remaining stable through N_{sub}=25. The locality assumption underlying subgraph extraction thus holds well for power grids, but may weaken for graphs with strong long-range dependencies.

F. Sensitivity and Convergence

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$ (1.012–1.013). Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the predicted accuracy-robustness tradeoff: $c = 0.5$ gives $r_v = 0.090$ (72.1% acc); $c = 0.9$ gives $r_v = 0.051$ (80.6% acc).

Convergence. 100% convergence across all 9 datasets and 10 seeds. Pseudospectral index $\eta = 1.02$–1.28, confirming mild non-normality and validating the κ-based bounds.

G. Defense-Informed Edge Protection

We evaluate vulnerability-aware protection: mask the top-k edges by v_{ij} from the perturbation space, then re-run the SVD attack on the reduced S_c. On Cora (IGNN, $N = 50$, 10 seeds): masking the top-5 edges reduces SVD attack damage by 42±8% vs. 11±6% for random 5-edge masking. At top-10: $61 \pm 7\%$ vs. $18 \pm 9\%$. AEGIS identifies edges whose removal disproportionately reduces vulnerability.

TABLE V: S_c framework applied to 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall rank correlation of continuous S_c vulnerability scores vs. discrete single-edge-removal ground truth (brute-force). Test Acc: full-graph accuracy. GAT†: edge-weighted variant for continuous differentiability (see text).

Model	Tight.	AtkAdv	τ	Test Acc
IGNN (equil.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}$x	$+.32_{\pm.04}$	$77.5_{\pm1.7}\%$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}$x	$-.04_{\pm.03}$	$78.9_{\pm0.9}\%$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}$x	$+.49_{\pm.02}$	$78.3_{\pm1.8}\%$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}$x	$+.33_{\pm.03}$	$76.6_{\pm1.9}\%$
GAT-2†	$1.01_{\pm.01}$	$2.1_{\pm0.1}$x	$+.54_{\pm.06}$	$77.8_{\pm1.2}\%$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}$x	$+.22_{\pm.10}$	$77.5_{\pm1.5}\%$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}$x	$+.35_{\pm.01}$	$82.2_{\pm0.6}\%$

H. Extension to Explicit GNNs

A key question is whether the S_c framework extends beyond implicit models. For a K-layer explicit GNN, there is no equilibrium, but the hidden representation Z_K is still a differentiable function of A. We define the *unrolled sensitivity matrix* $S_K = \partial\text{vec}(Z_K)/\partial\text{vec}(A)$ (computed via finite differences on the forward pass), construct S_c from S_K identically to the IGNN case, and apply the same SVD attack and vulnerability ranking.

table V applies the S_c framework to six explicit GNN architectures on Cora. All six achieve near-perfect first-order tightness (0.99–1.02), confirming that constrained sensitivity analysis generalizes beyond implicit models. Attack advantage ranges from 2.0× (SAGE) to 7.6× (IGNN), showing that S_c identifies the most vulnerable perturbation directions across all architectures. Test accuracy ranges from 76.6% (GIN) to 82.2% (APPNP), competitive with deep residual architectures such as GCNII [35]. GAT† and GCN-4 produce the strongest vulnerability rankings (τ = +0.54 and +0.49). APPNP achieves the best accuracy-vulnerability balance (τ = +0.35, 82.2% accuracy).

GAT requires an edge-weighted formulation where $\hat{A}$ values modulate attention weights continuously; standard GAT uses A as a binary mask (zeroing non-edges), so $\partial Z/\partial A_{ij} = 0$ for all existing edges. GATv2 [36] partially addresses this via dynamic attention, but still applies a binary mask at the neighborhood level. Our edge-weighted variant multiplies attention coefficients by $\hat{A}_{ij}$, restoring continuous differentiability. With this modification, GAT achieves tightness 0.99, attack advantage 4.4×, and $\tau = +0.36$.

Continuous-to-discrete transfer. The τ column in table V directly measures whether continuous S_c vulnerability scores predict discrete edge-removal impact. The brute-force ground truth removes each edge entirely (a discrete, non-infinitesimal perturbation) and re-evaluates the model; S_c scores are derived from continuous first-order sensitivity. Positive τ across 6 of 7 architectures (+0.22 to +0.54) demonstrates that continuous first-order analysis is a reliable proxy for discrete structural vulnerability, despite the perturbation-model mismatch. The one exception is GCN-2 ($\tau = -0.04$), likely due to its shallow depth limiting sensitivity differentiation.

Comparison with gradient attribution. Against the same brute-force discrete ground truth, gradient-based edge attribution [37], [38] produces consistently negative τ (-0.07, -0.17, -0.21 for IGNN, GCN-2, GIN-2 respectively). Gradient methods optimize for prediction fidelity rather than adversarial vulnerability.

Proposition 4 bound tightness. The ratio (Prop. 4 bound)$/\sigma_1(S_K)$ ranges from 1.4× (SAGE-2) to 5.9× (APPNP), with 2-layer models achieving tighter ratios (GCN-2: 1.8×, GIN-2: 2.1×) than deeper ones (GCN-4: 4.2×, GAT-2: 3.7×). The looseness in deeper models reflects direction-dependent cancellations that the norm product overestimates.

Degree-weighted S_c. Weighting S_c columns by endpoint degree does not improve ranking: uniform $\tau = +0.042$, degree-weighted $+0.037$, inverse-degree -0.008 (GCN-2, 10 seeds). High-degree nodes already contribute larger sensitivity columns via aggregation.

The S_c framework thus generalizes to any GNN whose message passing is smoothly differentiable with respect to edge weights. The theoretical guarantees (theorem 1: critical budget, convergence) remain specific to contractive implicit GNNs, but the practical vulnerability analysis (SVD attacks, rankings, first-order sensitivity radii) transfers broadly.

VI. Case Study: Power Flow Contingency

The experiments above validate AEGIS on classification benchmarks. We now test whether the vulnerability spectrum captures domain-specific structure by applying it to AC power flow, where the spectrum recovers a classical engineering problem: N-1 contingency analysis. Recent work has applied GNNs to power flow computation [39], [40] and contingency screening [3]; AEGIS approaches the same problem from the vulnerability-analysis perspective.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage and flow deviations [41]. This brute-force procedure scales as $O(|E|)$ full power flow solves, which is expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|[S_c]_{:,k}\|_2$ computes an analogous quantity via a single IFT analysis: edge perturbation maps to line trips, the vulnerability spectrum maps to contingency severity, and critical budget $\varepsilon_{\text{crit}}$ maps to stability margin. The correspondence is approximate (continuous first-order vs. discrete removal), analogous to how PTDF/LODF [41] linearize the effect of line trips.

B. Experimental Setup

We train ContractiveGCN-PF, an IGNN variant for AC power flow, on four standard IEEE test cases [31]. Training

TABLE VI: AEGIS on IEEE power flow benchmarks (10 seeds). Model quality (per-unit RMSE): $|V|$ = voltage magnitude, θ = angle, ΔS = power-balance residual per bus. τ: Kendall rank correlation vs. brute-force N-1 contingency. P@10: top-10 precision.

			Model (p.u.)			AEGIS	
Case	N	$\lvert E\rvert$	$\lvert V\rvert$	θ	ΔS	τ	P@10
case14	14	20	.007	.020	.106	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	.042	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	.033	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	.042	$+.62_{\pm.11}$	$.81_{\pm.10}$

data is generated via PandaPower's [32] full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: bus type indicators (slack, PV), active and reactive power injections (P, Q), and voltage magnitude $|V|$. Unlike graph benchmarks, power flow grids are small enough (14–118 buses) to analyze without subgraph extraction.

Limitation. Training data covers uniform load scaling (2,000 samples) but not seasonal or generator-outage variation.

C. Results

table VI reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

Key observations. First, tightness transfers to the power flow domain (≈ 1.00), confirming that the first-order prediction is not domain-specific. Second, ranking correlation ($\tau = 0.37$–0.67) is moderate but the operationally relevant P@10 is 0.66–0.81, correctly identifying 7–8 of the top 10 critical lines.

Observation: approximate physics from equilibrium structure. The model is trained with voltage MSE alone, without any power-balance penalty. Yet the per-bus residual $\Delta S = 0.03$–0.11 p.u. (table VI), meaning predicted voltages approximately satisfy Kirchhoff's laws. This is consistent with prior observations that equilibrium models absorb structural constraints from training data [16]: the fixed-point condition $Z^* = F(Z^*, A)$ forces self-consistency with the grid topology in A. Perturbations that disrupt this self-consistency are precisely those that break the power balance, connecting structural vulnerability to physical contingency. We note, however, that the residuals ($\Delta S = 0.03$–0.11 p.u.) are substantial in power engineering terms, and a controlled comparison against explicit GNN architectures on the same task would be needed to confirm this is an architectural rather than data-driven effect.

Baseline comparison. LODF (industry-standard DC screening [41]) achieves $\tau = 0.44$–0.58; AEGIS outperforms on larger grids ($\tau = 0.62$–0.67 on case57/118) by capturing nonlinear AC effects. LODF requires line reactances; AEGIS requires only the trained model.

Practical positioning. For case118, AEGIS ranks in 2.3s (single IFT) vs. 179 full AC solves for brute-force N-1, achieving P@10 = 0.81. Binary adjacency outperforms admittance-weighted (P@10 = 0.81 vs. 0.27) because N-1 contingency is a discrete event better modeled by uniform sensitivity.

Operational caveat. AEGIS is a screening layer, not a standalone contingency tool ($\tau = 0.37$–0.67 is insufficient for direct operational use).

VII. Related Work

We position AEGIS against four research threads.

Adversarial attacks on graph structure. Nettack [1] and Mettack [4] demonstrated structural vulnerability via targeted and global poisoning using GCN [33] surrogates. Subsequent work explored RL-based [2], spectral [42], and bilevel [43] formulations; Wu et al. [44] survey the landscape. These methods use surrogate gradients or heuristic search. AEGIS instead computes the closed-form optimal attack direction via SVD of S_c.

Certified robustness and defense. Convex relaxations [8], [7], randomized smoothing [5], [6], and deterministic certificates [45], [46] provide robustness guarantees but do not identify *which* edges are vulnerable. Empirical defenses [10], [12], [11] harden models without formal vulnerability maps. AEGIS complements both by providing per-edge rankings and per-node radii that reveal the attack surface geometry (section V-B).

Implicit networks and equilibrium analysis. The DEQ framework [16] introduced fixed-point models; IGNN [14] enforces unique equilibria via spectral normalization [18]; EIGNN [47] improves efficiency. For robustness, monotone operator networks [48] provide built-in Lipschitz bounds, and El Ghaoui et al. [24] analyze well-posedness. All address input sensitivity $\|\partial z^*/\partial x\|$, not structural sensitivity $\|\partial Z/\partial A\|$. AEGIS addresses structural sensitivity via the resolvent $(I - J_z)^{-1} J_A$ for implicit models (theorem 1) and the unrolled Jacobian S_K for explicit ones (theorem 4).

Sensitivity analysis and explainability. AEGIS draws on matrix perturbation theory [26], [22]. GNN explanation methods (GNNExplainer [38], gradient-based [37]) optimize for *prediction fidelity*, not adversarial vulnerability. AEGIS's S_c is perturbation-space optimal by construction (SVD) and provides quantitative sensitivity radii. Our case study (section VI) connects to GNN-based power flow [39], [40] and contingency screening [3].

VIII. Conclusion

AEGIS provides a principled pre-deployment diagnostic for GNNs on safety-critical graphs: from a single S_c computation, practitioners obtain the optimal attack direction, per-edge vulnerability rankings, and per-node tolerance budgets. Predictions are tight across 7 architectures and 9 datasets (tightness 1.00 ± 0.01 at $\varepsilon = 0.01$), and the vulnerability spectrum transfers to power grid contingency analysis (P@10 = 0.66–0.81). The per-edge ranking directly informs defense design (section V-G).

Limitations. (1) First-order sensitivity radii are locally tight but not global certificates without second-order bounds. (2) Continuous edge-weight perturbations; discrete insertions/deletions are outside the formal guarantee, though continuous S_c rankings transfer well to discrete removal ($\tau = +0.22$–$+0.54$; table V). (3) Standard GAT requires edge-weighted modification for continuous sensitivity. (4) Power flow $\tau = 0.37$–0.67 requires verification for operational use [41]. (5) The matrix-free pipeline's Neumann series requires $\kappa < 1$; for explicit GNNs without contractivity, the dense path ($N \leq 200$) or finite-difference columns are used. (6) IGNN accuracy (77.5% on Cora) lags behind state-of-the-art explicit GNNs ($\sim$82% APPNP); this reflects the accuracy–guarantee tradeoff inherent in contractive architectures, not a limitation of AEGIS itself. The S_c framework applies identically to higher-accuracy explicit models, albeit without the formal regime characterization (theorem 1).

Ethical considerations. AEGIS identifies optimal attack directions and structurally vulnerable edges, which could be misused to attack deployed GNNs in safety-critical domains. We argue the diagnostic value outweighs this risk: defenders must understand their system's attack surface to protect it, and the same vulnerability map directly informs defense design (section V-G). We will release code with responsible-use guidelines, consistent with established practice in adversarial ML [44].

Future work. Second-order bounds for formal certificates; discrete perturbation models; integration with defense mechanisms [12], [11] for vulnerability-aware training; extending full-graph analysis beyond $N \approx 5{,}000$ via distributed JVP computation.

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